Neeraj Kumar Kanchani

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SUMMARY

Robotics software engineer building tested Python and C++ systems for distributed edge devices, real-time perception, state estimation, and bounded control in ROS 2 and Gazebo.

TECHNICAL SKILLS

Languages: Python, C++, C, MATLAB | **Robotics:** ROS 2 Jazzy, Gazebo Harmonic, state estimation, control systems, multi-robot systems
Perception: YOLOv8, OpenCV, monocular geometry | **Systems:** Linux, Docker, V4L2, UNIX sockets, GitHub Actions, CI/CD, automated testing

TECHNICAL PROJECTS

Vision-Only Leader-Follower Warehouse Coordination github.com/justneeraj12/ee616-vision-leader-follower
ROS 2 Jazzy, Gazebo Harmonic, Python, YOLOv8n, OpenCV, wheel odometry, Docker

- Validated string stability across 12 three-follower chain runs and 18 visual-blackout/yaw-bias disturbance runs (0.1004 m, 0.0739 m, and 0.0523 m worst-case RMSE; zero collisions); designed deterministic TRACK, PREDICT, and SAFE STOP modes using monocular pinhole geometry, wheel odometry, and bounded range-and-bearing control.
- Diagnosed YOLOv8n RGB/BGR tensor-channel mismatches and stale-frame timestamps, reaching 100% valid detections, 0.0323 m range RMSE, 0.1569 deg bearing MAE, and 9.02 ms p95 inference latency.
- Architected a containerized ROS 2 Jazzy/Gazebo Harmonic convoy across isolated namespaces with 5 packages and 61 tests; on an RTX 3050 Ti (4 GB), measured 23.09 ms p95 camera-to-measurement, 48.35 ms p95 measurement-to-control, 642 MiB peak VRAM, and 0.9996 minimum real-time factor.

CPU-First Local Face Authentication for Linux github.com/justneeraj12/face-unlock-linux
C++17, C, PAM, Qt 6, OpenCV DNN, V4L2, UNIX sockets, libsodium, CMake

- Built a CPU-only ONNX pipeline with YuNet 5-landmark detection and SFace 112x112 alignment/embeddings, measuring 6.6 ms detection and 10.1 ms p50/12.6 ms p95 embedding latency without CUDA.
- Engineered a C/C++17 systems layer with a minimal PAM bridge, on-demand V4L2 capture, credential-checked UNIX socket IPC, libsodium-encrypted profiles, bounded timeouts, and password/PIN fallback under fail-closed authentication.

Nocturne Operations Console for Hyprland github.com/justneeraj12/nocturne-hyprland
C++, Qt 6, Qt Quick, LayerShellQt, Wayland, DBus, Linux IPC

- Built a native C++/Qt 6 and Qt Quick control surface using LayerShellQt on Wayland, with DBus and Linux IPC integrations for standard desktop services.
- Added deterministic recovery and upgrade verification with preflight snapshots, read-only diagnostics, and confirmation-gated rollback tested in clean Ubuntu and Arch CI containers.

EXPERIENCE

Graduate Grader, Digital Signal Processing (EE 501) Aug 2026 - Dec 2026
Clarkson University, Dr. Banavar / Department of Electrical and Computer Engineering | Potsdam, NY

- Evaluated graduate DSP problem sets and MATLAB/Python simulations covering discrete-time systems, Fourier analysis, Parseval's theorem, sampling/aliasing, and filter stability; authored master solution keys and diagnostic feedback.

Graduate Research Assistant, Theoretical Systems Sep 2025 - Jan 2026
Clarkson University, Dr. Banavar's Research Group | Potsdam, NY

- Benchmarked Gauss-Newton against linearized least squares for nonlinear TOA localization and reduced simulated RSSI error by 50% through GDOP-based anchor placement across a 200-node wireless network.

Undergraduate Research Assistant, UAVs and Robotics Jan 2023 - Jun 2024
Cambridge Institute of Technology, Dr. Antony's Lab | Bengaluru, India

- Integrated perception and embedded control across 2 UAV/UUV prototypes, achieving 92% real-world obstacle-avoidance success and reducing OpenCV tracking latency by 30 ms.

Student Intern, Drone Technology Feb 2023 - Jun 2023
Avian Aerospace | Bengaluru, India

- Calibrated ArduPilot flight controllers across 4 drone prototypes, reducing mid-flight telemetry errors by 12% through PID tuning and flight-log analysis.

EDUCATION

Clarkson University Potsdam, NY
M.S. in Electrical and Computer Engineering | Expected December 2026 | *GPA: 3.5/4.0*

Visvesvaraya Technological University Bengaluru, India
B.E. in Artificial Intelligence and Machine Learning | July 2025 | *Honors: First Class with Distinction*

UNDERGRADUATE LEADERSHIP

Cambridge Institute of Technology Bengaluru, India
Chair, IEEE Robotics and Automation Society College Chapter; President, College Robotics Club